

CSCI 4485

Software Engineering

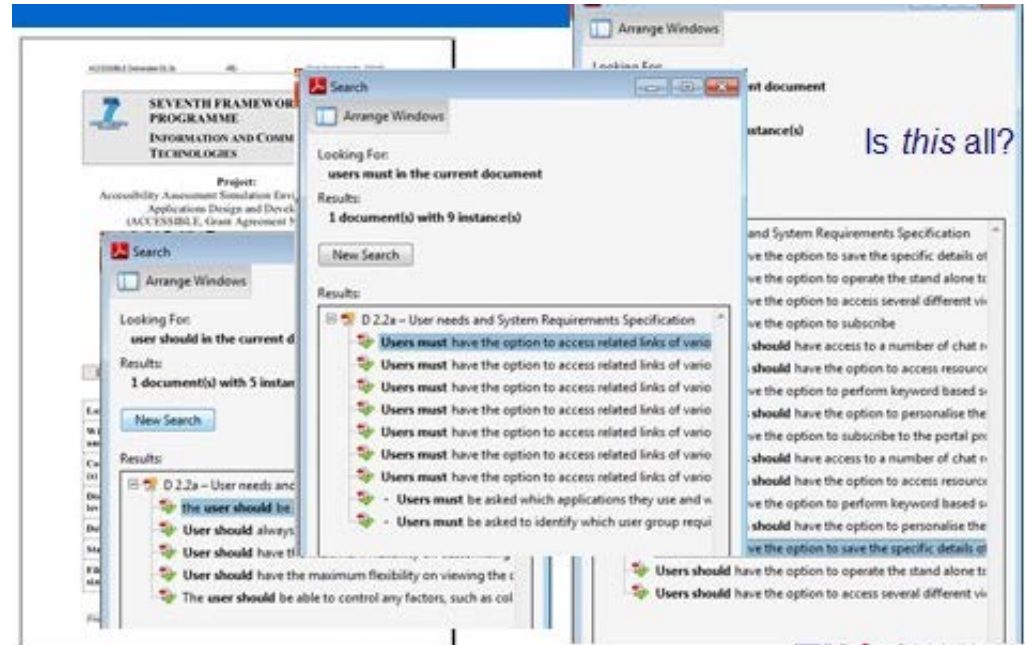
Use Cases

Outline

- What is a use case?
- Terminology
- Styles of use cases
- Steps for creating a use case

We need a better way

- **Goal:** specify how different parties (users, administrators, external systems, ...) can interact with our system
- *Recall: unambiguous, realistic, verifiable and etc.*
- Plain-text does not work
- A more **structured** way is needed



What is a use case?

A written description of the user's interaction with the software product to accomplish a goal.

- It is an example behavior of the system.
- Written from actor's point of view, not the system's.

Use cases capture functional requirements of a system.

Benefits of use cases

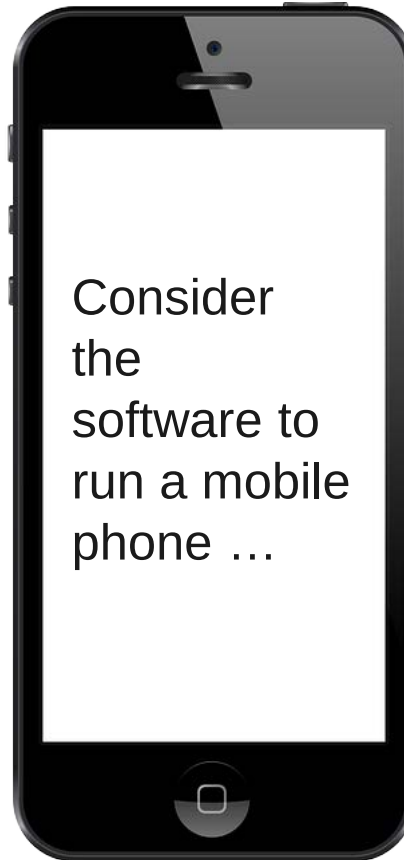
- Establish an understanding between the customer and the system developers of the requirements (**success scenarios**)
- Alert developers of problematic situations, error cases to test (**extension scenarios**)
- Capture a level of functionality to plan around (**list of goals**)

Use cases versus internal features

Use cases

- call someone
- receive a call
- send a message
- memorize a number

Point of view: user



Internal functions

- transmit / receive data
- energy (battery)
- user I/O (display)
- phone-book mgmt.

Point of view: developer

Use cases and requirements

- Special deals may not run longer than 6 months.
- Customers only become preferred after 1 year.
- A customer has one and only one sales contact.
- Database response time is less than 2 seconds.
- Web site uptime requirement is 99.8%.
- Number of simultaneous users will be 200 max.

Which of these requirements should be represented directly in a use case?

None! These are properties but not user-driven *behaviors* of the system, so the use cases wouldn't mention them.

Styles of use cases

Use case diagram

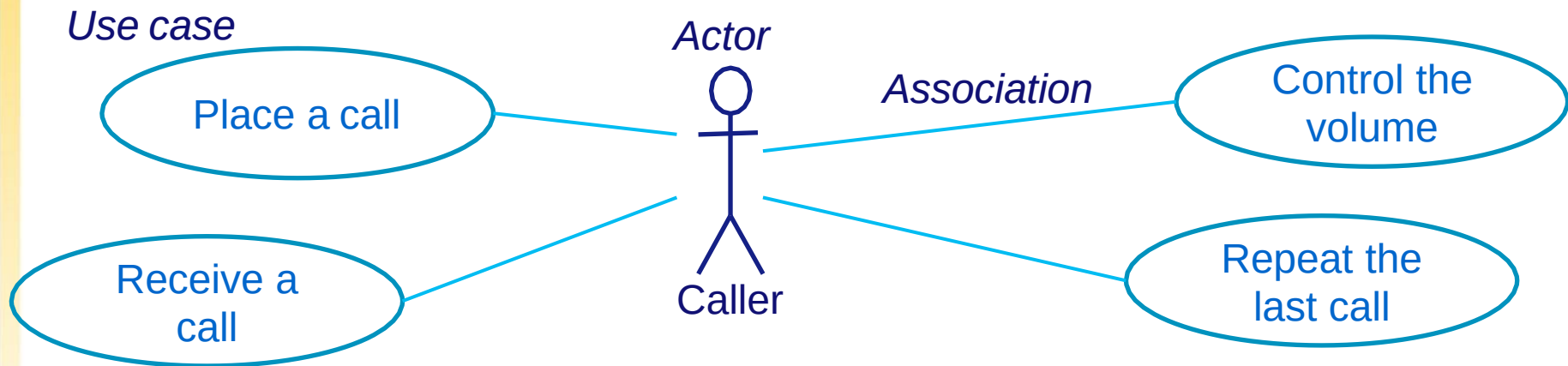
- in UML, the Unified Modeling Language

Use case specification

- a multi-part structured description

Simple example

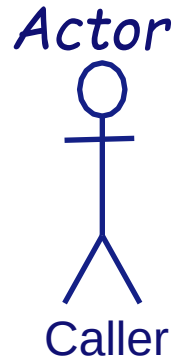
- What can you do with a telephone?



Basic use case diagram

What is an actor?

- A class of entities (human or computer), falling beyond the system boundaries, interacting with the system



- Note:
 - Actor can be an external software system
 - might have multiple actors in a use case:
caller/callee
 - One user might be represented by multiple actors
doctor can be a patient

Types of actors

- A primary actor
 - ✓ acts on the system
 - ✓ initiates an interaction with the system
 - ✓ uses the system to fulfill his/her goal
- A secondary actor
 - ✓ is acted on/invoked/used by the system
 - ✓ helps the system to fulfill its goal

Use Cases

- A **use case** tells a story of actors using a system.
 - “Rent Videos”
- Use cases:
 - Help to focus on achieving well-defined *user goals*.
 - Vary widely in their level of detail.
- Usually, a large **end-to-end** process comprising **several (inter)actions**

Place a call

Terminology

- **Actor:** an entity that acts on the system
 - Person, external hardware (like a timer), or another system.
- **Goal:** desired outcome of the primary actor

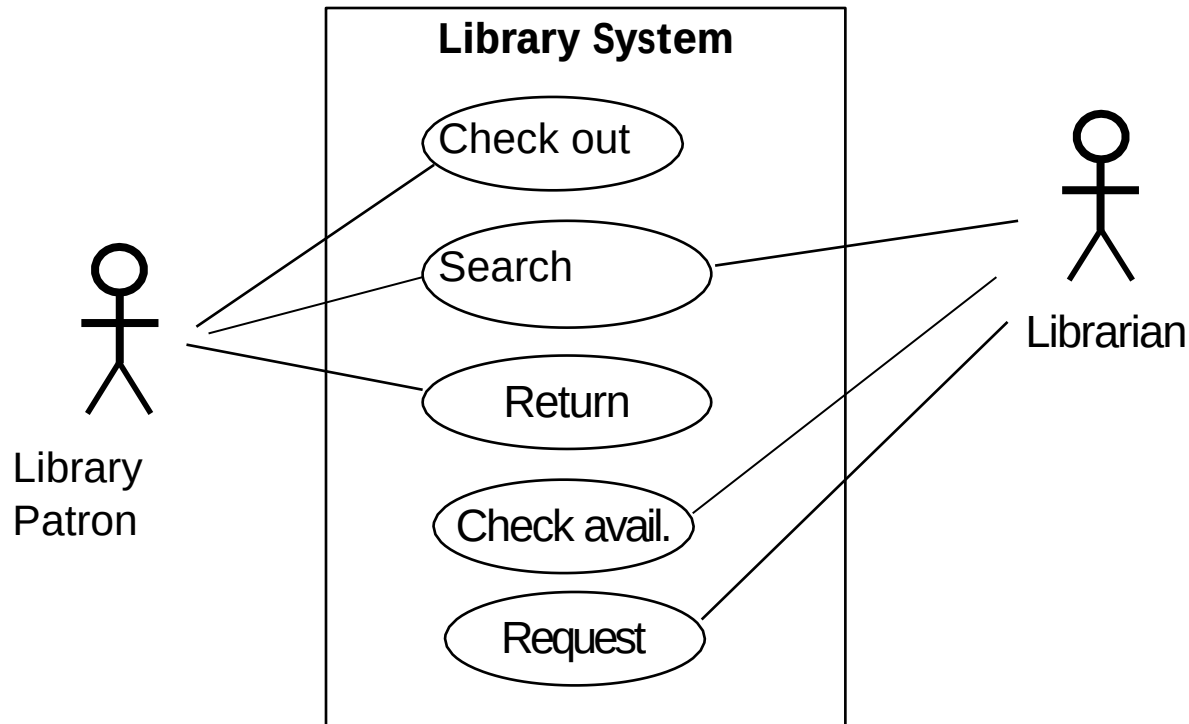
Use cases are always initiated by actors and describe the **flow of events** that these actors are involved in.

Actor-goal lists: function content of the system

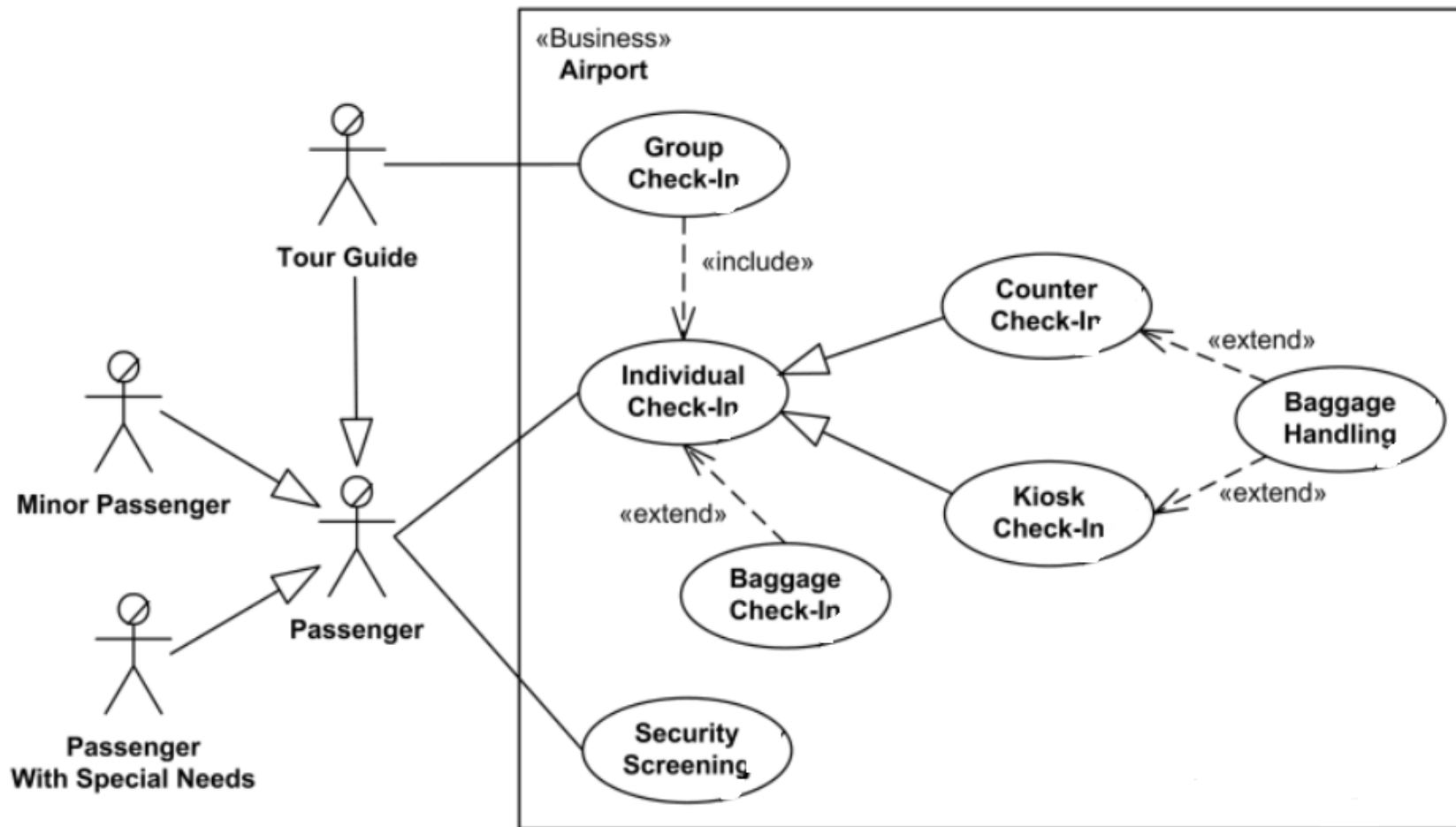
Actor	Goal
Library Patron	Search for a book
	Check out a book
	Return a book
Librarian	Search for a book
	Check availability
	Request a book from another library

It can be useful to create a list or table of primary actors and their "goals" (use cases they start). The diagram will then capture this material.

Use case summary diagrams

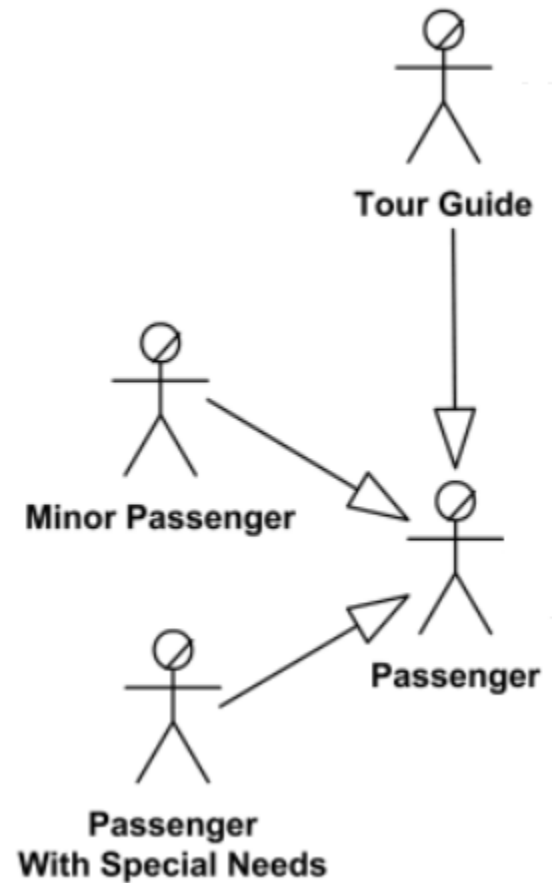


A more complex example



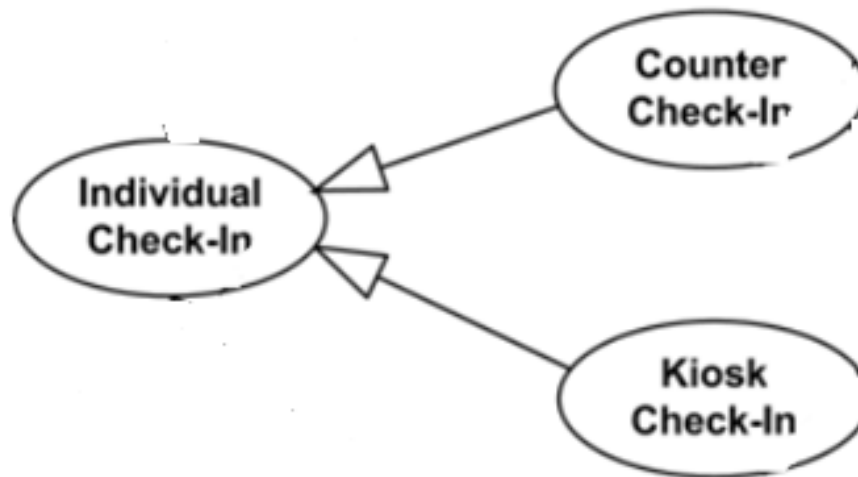
Generalization

- Some Passengers are Tour Guides, but all Tour Guides are Passengers
 - Tour Guide is a special Passenger.
- Also, Minor Passenger is a special passenger.

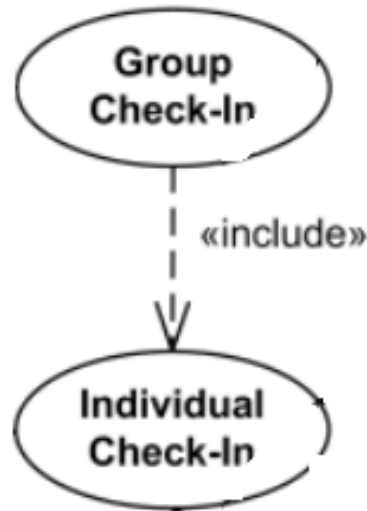


What about use cases?

- Counter check-in is a special case of individual check-in.
- Kiosk check-in is another special case of individual check-in.

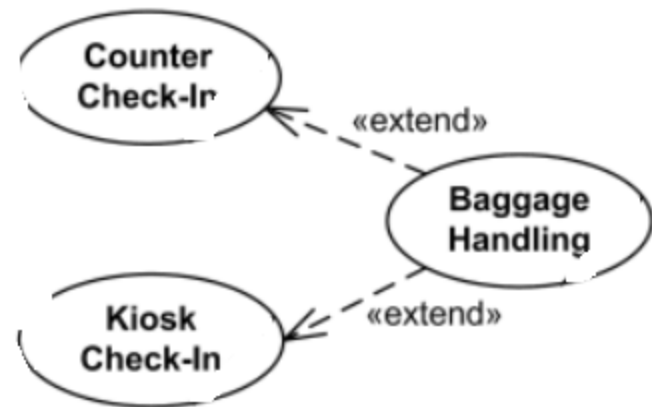
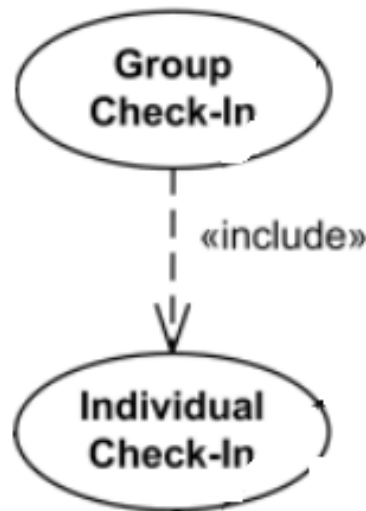


Including a use case in another use case

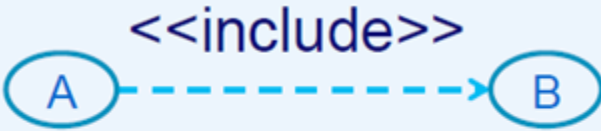
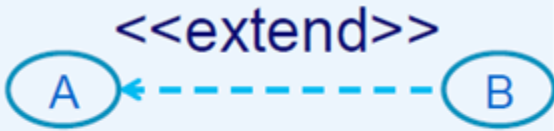


Formally

- **Dependency:** A use case depends on another use case for realizing its goal.
 - The target use case is *always* used by the source.
- What if the use case is used only sometimes?
 - Alternatives in the use case
 - **Extension**



Tips

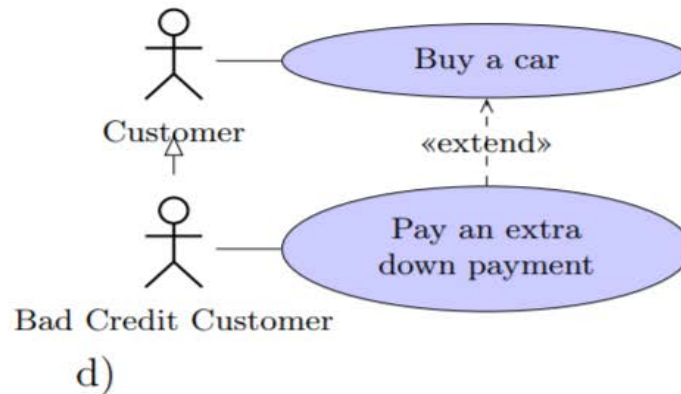
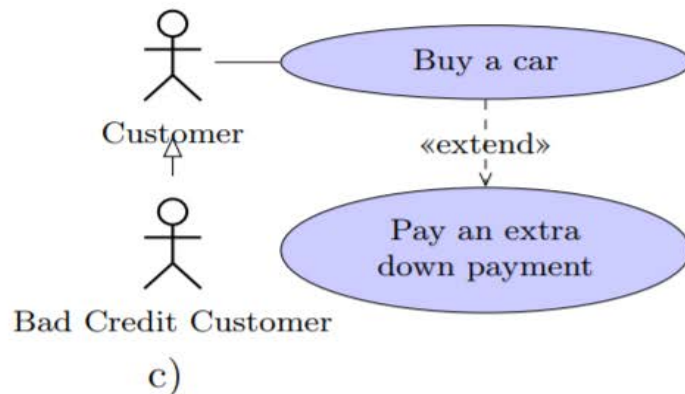
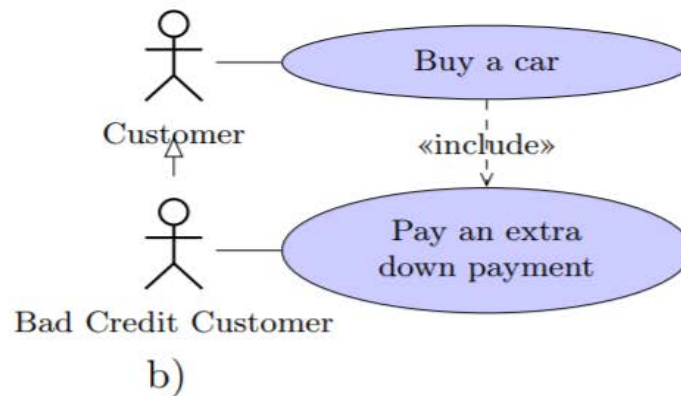
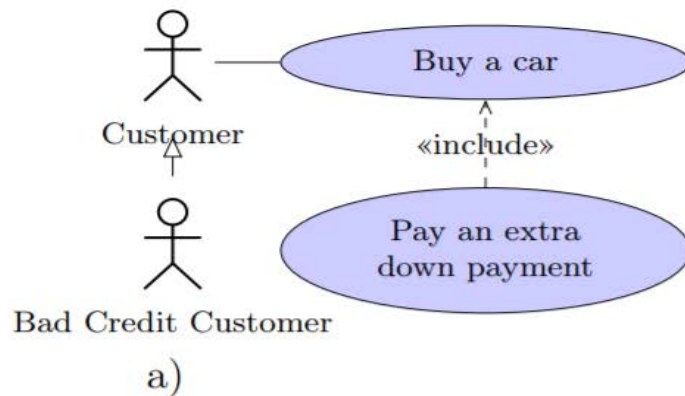
		
Read	A includes B	B extends A
A can operate without B	no	yes

Exercises

Respond at [PollEv.com/zhengli751](https://poll-ev.com/zhengli751)

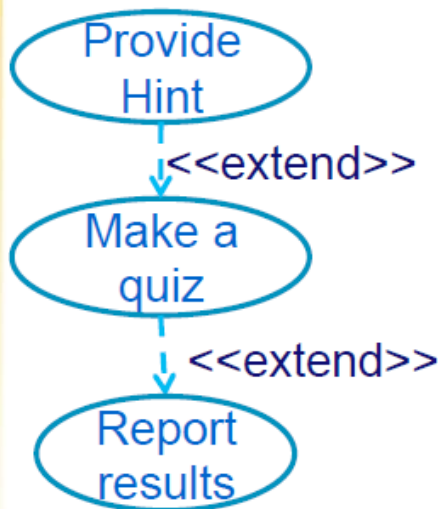
Text **ZHENGLI751** to **22333** once to join, then **A, B, C, D**

- “Customers of the garage can buy cars. Customers with a bad credit should pay an extra down payment”. Which of the following diagrams represent this description?

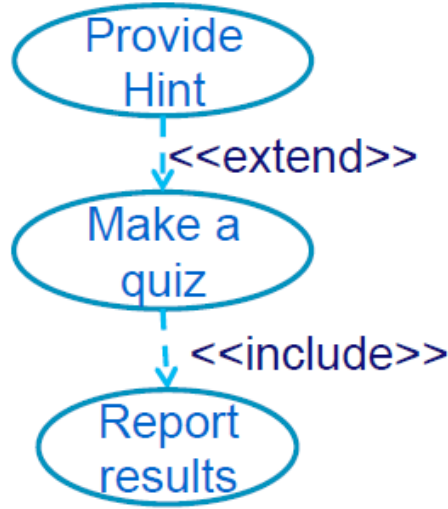


Which model is correct?

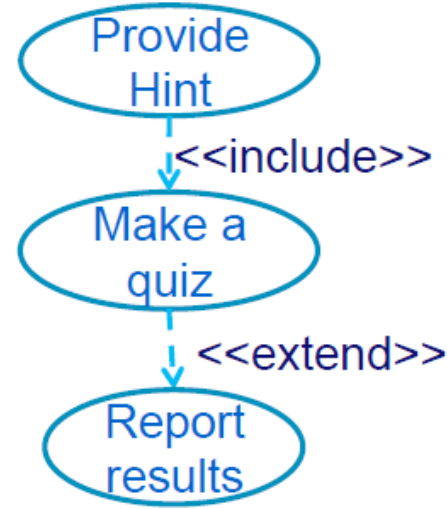
Digital learning environment supports students' learning through quizzes, report will be generated after the quiz. Given your knowledge of this kind of systems which of the following models would be correct?



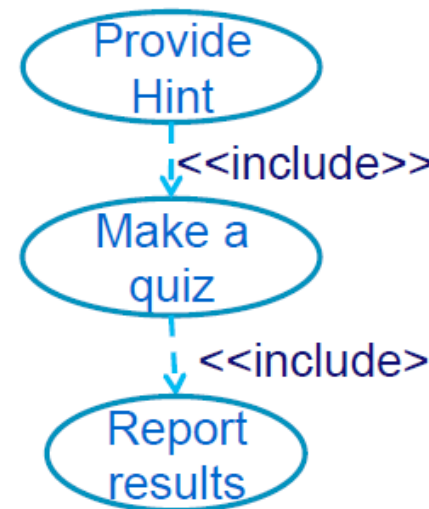
A



B



C

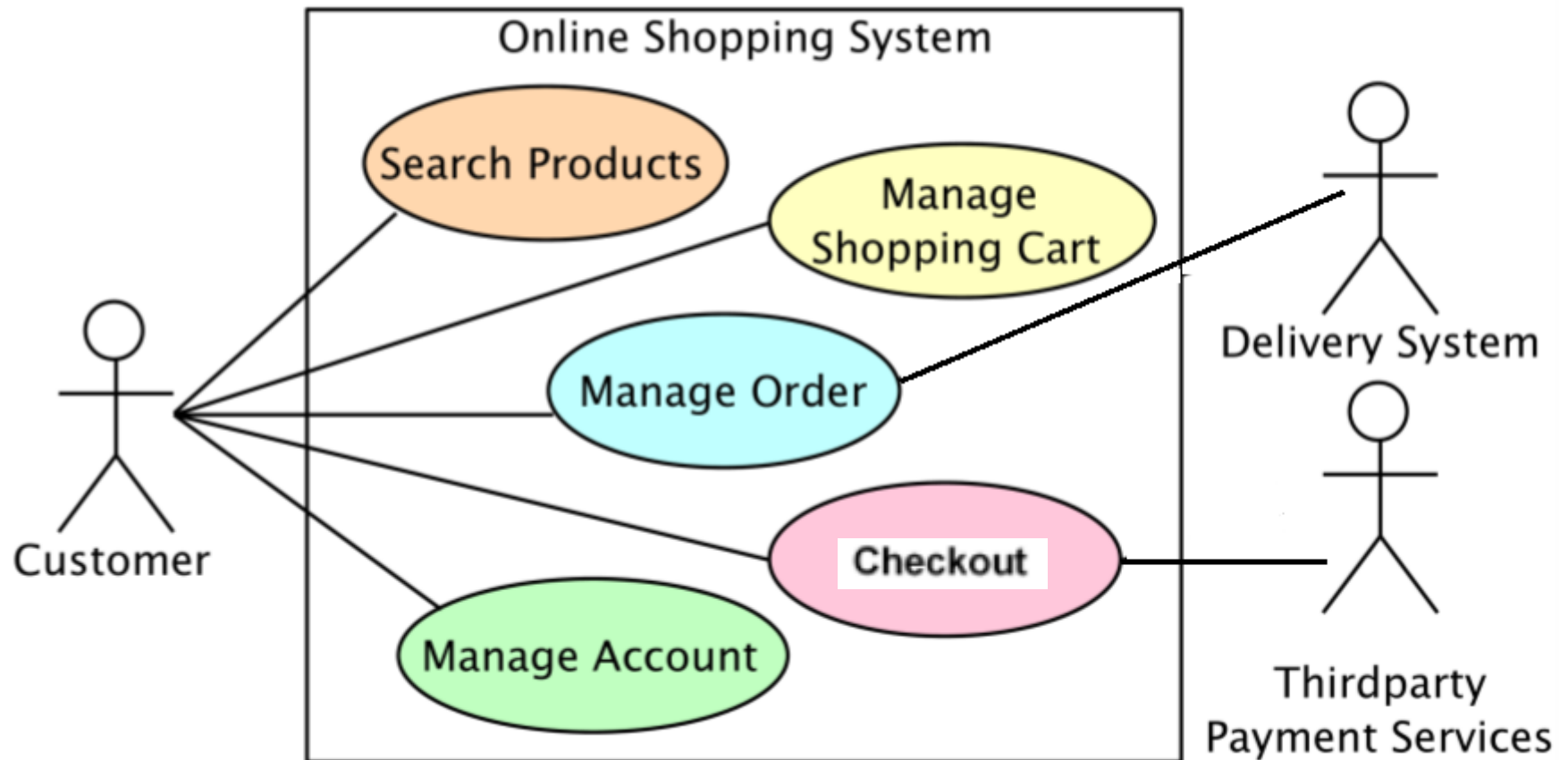


D

Respond at [PollEv.com/zhengli751](https://www.poll-ev.com/zhengli751)

Text **ZHENGLI751** to **22333** once to join, then **A, B, C, D**

Case Study: Online Shopping System



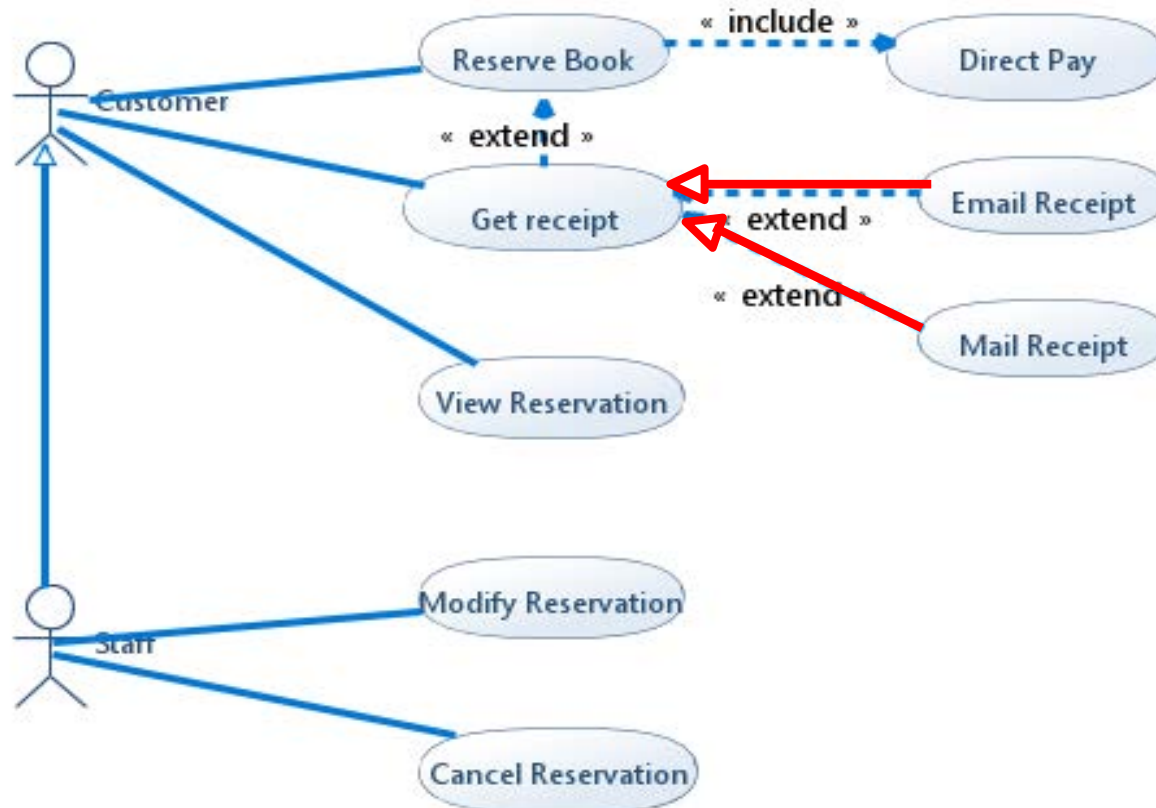
Top level use cases

Group Exercise (20 minutes)

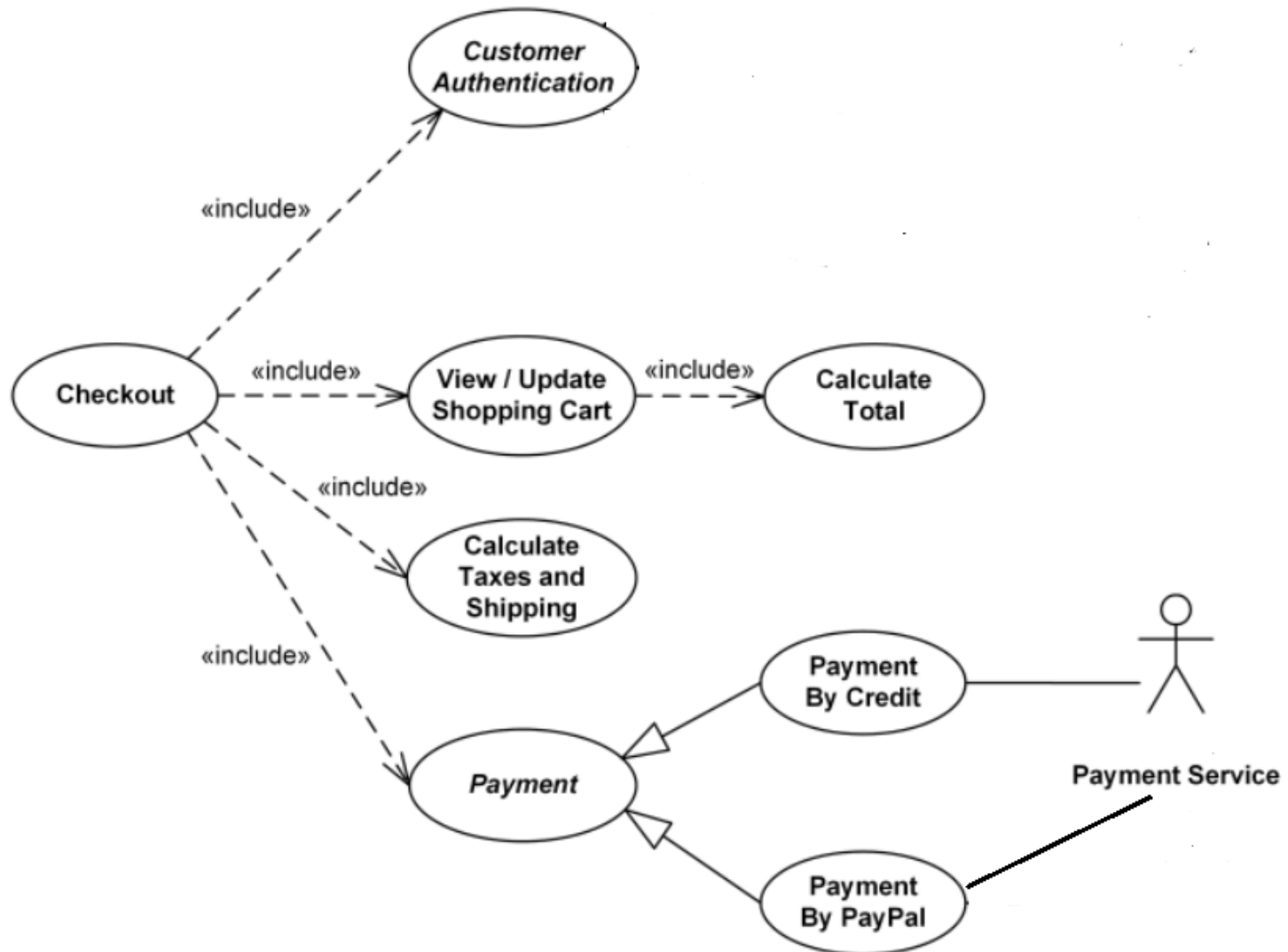
- **Please draw a Use Case Diagram for the following system:**

Consider an online shopping system for a bookstore. Books can be reserved by customers on the website and the customer must directly pay for the reserved books through the website. The customer has the option to request the reservation receipts emailed or mailed to them. The customers can view but can NOT modify or cancel any book reservation. The store staff can view, modify and cancel the book reservations.

Bug fix



Use case diagram - checkout



Styles of use cases

Use case diagram

- in UML, the Unified Modeling Language

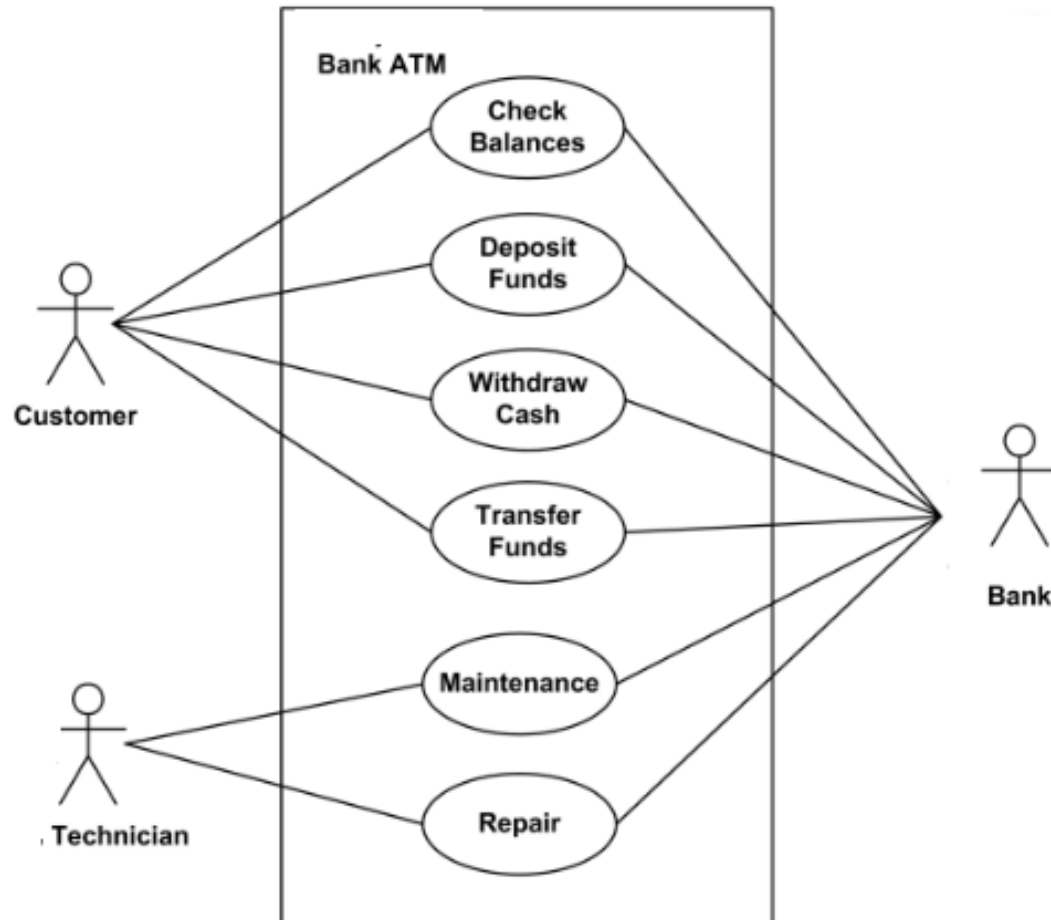
Use case specification

- a multi-part structured description

Use Case Specification

Use Case UC-#:	Name / Identifier	[verb phrase]
Related Requirements:	List of the requirements that are addressed by this use case	
Initiating Actor:	Actor who initiates interaction with the system to accomplish a goal	
Actor's Goal:	Informal description of the initiating actor's goal	
Participating Actors:	Actors that will help achieve the goal or need to know about the outcome	
Preconditions:	What is assumed about the state of the system before the interaction starts	
Postconditions:	What are the results after the goal is achieved; i.e., what must be true about the system at the time the execution of this use case is successfully completed	
Flow of Events for Main Success Scenario:		
→	1.	The initiating actor delivers an action or stimulus to the system (the arrow indicates the direction of interaction, to- or from the system)
←	2.	The system's reaction or response to the stimulus; the system can also send a message to a participating actor, if any
→	3.	...
Flow of Events for Extensions (Alternate Scenarios):		
What could go wrong? List the exceptions to the routine and describe how they are handled		
→	1a.	For example, actor enters invalid data
←	2a.	For example, power outage, network failure, or requested data unavailable
		...
The arrows on the left indicate the direction of interaction: → Actor's action; ← System's reaction		

Bank ATM System



Use Case: Withdraw Cash

Use Case UC-1: Withdraw

Related Requirem'ts: REQ1, REQ3, REQ4, and REQ5

Primary Actor: Customer

Actor's Goal: To withdraw money from one of Customer's account..

Secondary Actors: Banking System

Preconditions:

- The ATM is operational.
- The bank customer has a card to insert into the ATM.

Postconditions: The bank customer has received their cash (and optionally a receipt). And the bank has debited the customer's bank account and recorded details of the transaction.

Flow of Events for Main Success Scenario:

- 1. The customer enters their card into the ATM.
2. The ATM verifies that the card is a valid bankcard.
- ← 3. The ATM requests a PIN code.
- 4. The customer enters their PIN code.
5. The ATM validates the bankcard against the PIN code.

- ← 6. The ATM presents service options including “Withdraw”.
- 7. The customer chooses “Withdraw”.
- 8. The ATM verifies sufficient funds in the customer’s bank account.
- ← 9 The ATM presents options for amounts.
- 10. The customer selects an amount or enters an amount.
- 11. The ATM verifies that the customer is below withdrawing limits.
- 12. The ATM verifies that it has enough cash in its hopper.
- ← 13. The ATM debits the customer’s bank account.
- 14. The ATM returns the customer’s bankcard.
- 15. The customer takes their bankcard.
- ← 16. The ATM issues the customer’s cash.
- 17. The customer takes the cash.

Alternate Flows:

5A1: The user entered an invalid PIN.

- ← 1. The ATM will indicate that the wrong PIN has been entered.
- 2. The system will log and check number of tries.
- ← 3. The ATM will ask the customer to enter their PIN again.
- 4. The use case returns to step 4 and continues

... ..

Exercise

- Where in a use case for Borrower Loans a Copy would the following fit?

A new Loan object has been created and all data (references to a Borrower, a Copy and the Catalog and dates and times for the issue of the loan and when it is due) have been provided and saved and the collections of all loans for the Catalog and the relevant Borrower have been updated.

- A. As a pre-condition
- B. As a post-condition
- C. As an exception flow
- D. As an alternative flow

Traceability Matrix

Mapping: System requirements to Use cases

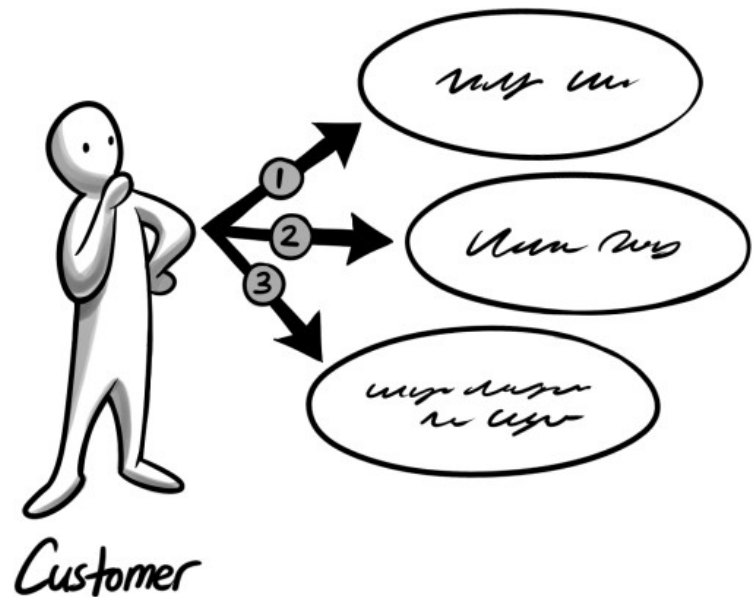
REQ1: Keep door locked and auto-lock
 REQ2: Lock when "LOCK" pressed
 REQ3: Unlock when valid key provided
 REQ4: Allow mistakes but prevent dictionary attacks
 REQ5: Maintain a history log
 REQ6: Adding/removing users at runtime
 REQ7: Configuring the device activation preferences
 REQ8: Inspecting the access history
 REQ9: Filing inquiries

UC1: Unlock
 UC2: Lock
 UC3: AddUser
 UC4: RemoveUser
 UC5: InspectAccessHistory
 UC6: SetDevicePrefs
 UC7: AuthenticateUser
 UC8: Login

Req't	PW	UC1	UC2	UC3	UC4	UC5	UC6	UC7	UC8
REQ1	5	X	X						
REQ2	2		X						
REQ3	5	X						X	
REQ4	4	X						X	
REQ5	2	X	X						
REQ6	1			X	X				X
REQ7	2						X		X
REQ8	1					X			X
REQ9	1					X			X
Max PW		5	2	2	2	1	5	2	1
Total PW		15	3	2	2	3	9	2	3

Steps for creating a use case

1. Identify actors and goals
2. Write the main success scenario
3. List the alternatives



Pulling it all together: how much is enough?

You have to find a balance

- comprehensible vs. detailed
- graphics vs. explicit wording and tables
- short and timely vs. complete and late

Your balance may differ with each customer depending on your relationship and flexibility